

ANDROID APP DEVELOPMENT USING GENAI - MAHILA-SHAKTI UNNATI (MICRO-FINANCE)

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ECE

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INTRODUCTION

- Mahila-Shakti Unnati is an Android application developed using Kotlin and Jetpack Compose with the concept of Generative AI for micro-finance support. The main aim of this project is to help women manage financial activities easily by providing loan eligibility checking and basic financial guidance through a simple mobile application.
- Many women, especially in rural and low-income communities, face challenges in accessing financial services and understanding loan processes. This application provides a user-friendly platform where users can enter their income and loan details to quickly check eligibility.

OBJECTIVES

- To develop a simple and user-friendly Android application for micro-finance support.
- To help women check loan eligibility easily using income and loan details.
- To promote financial awareness and digital accessibility among women.
- To provide quick and accurate financial assistance through AI-based concepts.
- To implement modern Android development technologies like Kotlin and Jetpack Compose.

ABOUT THE PROJECT

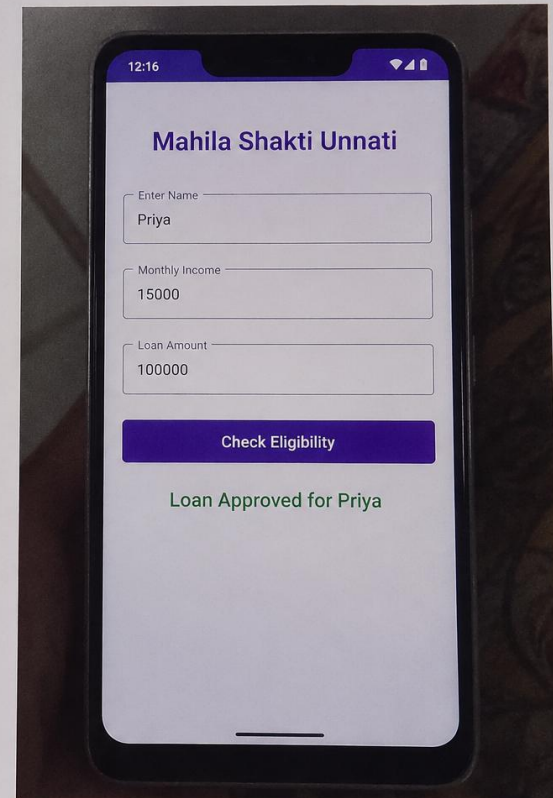
- Mahila-Shakti Unnati is an Android-based micro-finance application developed using Kotlin and Jetpack Compose with the support of Generative AI concepts. The project is designed to help women access basic financial services in a simple and digital way. The application allows users to enter their personal and financial details, such as monthly income and loan amount, to check loan eligibility instantly.
- The main purpose of the project is to promote financial awareness, women empowerment, and easy access to micro-finance solutions through mobile technology. The app provides a clean and user-friendly interface using Material Design 3 components and modern Android development practices. This project also demonstrates how AI-based concepts can be integrated into Android applications to create smart, efficient, and accessible financial assistance systems.

WORK DONE

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In this project, an Android application named Mahila-Shakti Unnati was developed using Kotlin and Jetpack Compose to provide micro-finance support for women. The following work has been completed:

- ✓ Developed an Android application using Kotlin and Jetpack Compose.
- ✓ Designed a simple and user-friendly interface for financial assistance.
- ✓ Created input fields for user name, monthly income, and loan amount.
- ✓ Implemented loan eligibility checking logic based on income details.
- ✓ Used Compose State Management (remember and mutableStateOf) to manage user inputs dynamically.
- ✓ Added result display functionality for loan approval or rejection.
- ✓ Applied Material Design 3 components for a modern UI design.
- ✓ Integrated the concept of Generative AI-based financial assistance into the application.



WORK DONE

TECHNOLOGIES USED



1. Kotlin

Programming language used to build the Android application.



2. Android Studio

IDE used for development, debugging and testing of the application.



3. Jetpack Compose

Modern UI toolkit used to build native Android UI declaratively.



4. Material Design 3

Design system used for building modern and consistent UI.



5. Android SDK

Used to run and test the application on Android devices/emulators.

```
package com.example.mypaplication

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            MahilaShaktiApp()
        }
    }
}

@Composable
fun MahilaShaktiApp() {
    var name by remember { mutableStateOf("") }
    var income by remember { mutableStateOf("") }
    var loan by remember { mutableStateOf("") }
    var result by remember { mutableStateOf("") }

    MaterialTheme {
        Column(modifier = Modifier.fillMaxSize().padding(20.dp)) {
            Text(text = "Mahila Shakti Unnati", style = MaterialTheme.typography.headlineMedium)
            Spacer(modifier = Modifier.height(20.dp))
            OutlinedTextField(value = name, onValueChange = { name = it },
                label = { Text("Enter Name") }, modifier = Modifier.fillMaxWidth())
            Spacer(modifier = Modifier.height(10.dp))
            OutlinedTextField(value = income, onValueChange = { income = it },
                label = { Text("Monthly Income") }, modifier = Modifier.fillMaxWidth())
            Spacer(modifier = Modifier.height(10.dp))
            OutlinedTextField(value = loan, onValueChange = { loan = it },
                label = { Text("Loan Amount") }, modifier = Modifier.fillMaxWidth())
            Spacer(modifier = Modifier.height(20.dp))
            Button(onClick = {
                result = if (income.isNotEmpty() && loan.isNotEmpty()) {
                    val incomeValue = income.toInt()
                    val loanValue = loan.toInt()
                    if (loanValue <= incomeValue * 10) {
                        "Loan Approved for $name"
                    } else {
                        "Loan Rejected"
                    }
                } else {
                    "Enter all details"
                }, modifier = Modifier.fillMaxWidth()) {
                Text("Check Eligibility")
            }
            Spacer(modifier = Modifier.height(20.dp))
            Text(text = result, style = MaterialTheme.typography.headlineSmall)
        }
    }
}
```

EXPECTED OUTCOME

- The expected outcome of this project is to provide a simple, user-friendly, and accessible Android application that helps women understand basic financial services and loan eligibility easily. The application aims to improve financial awareness, support women empowerment, and simplify micro-finance processes through digital technology. By using Kotlin, Jetpack Compose, and Generative AI concepts, the project is expected to deliver quick financial assistance, efficient loan checking, and a modern mobile experience for users.

CONCLUSION

- Mahila-Shakti Unnati is a simple and effective Android application developed using Kotlin and Jetpack Compose to support women through micro-finance services. The project successfully demonstrates how modern Android technologies and Generative AI concepts can be used to provide smart financial assistance and loan eligibility checking. The application promotes financial awareness, digital accessibility, and women empowerment through an easy-to-use interface. Overall, this project highlights the importance of technology in creating accessible and efficient financial solutions for society.

THANK YOU...